



Sustainable Technology

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The saddest aspect of life right now is that
science gathers knowledge faster than
society gathers wisdom.

~ Isaac Asimov



From Stones to Silicon

Ancient Foundations

❖ 2.5 million years ago

Stone tools, control of fire

❖ ~3500 BC

Invention of the wheel, bronze tools

Creations like aqueducts, levers, and pulleys demonstrated early engineering.



Cueva de las Manos

Leap towards better future

➤ **Industrial Age (1760–1840)**

Steam-powered factories enable mass production and urban growth.

➤ **Early Modern Age (19th–20th Century):**

Efficiency and industrial expansion prioritized over environmental impact.



Power House Mechanic by Hine

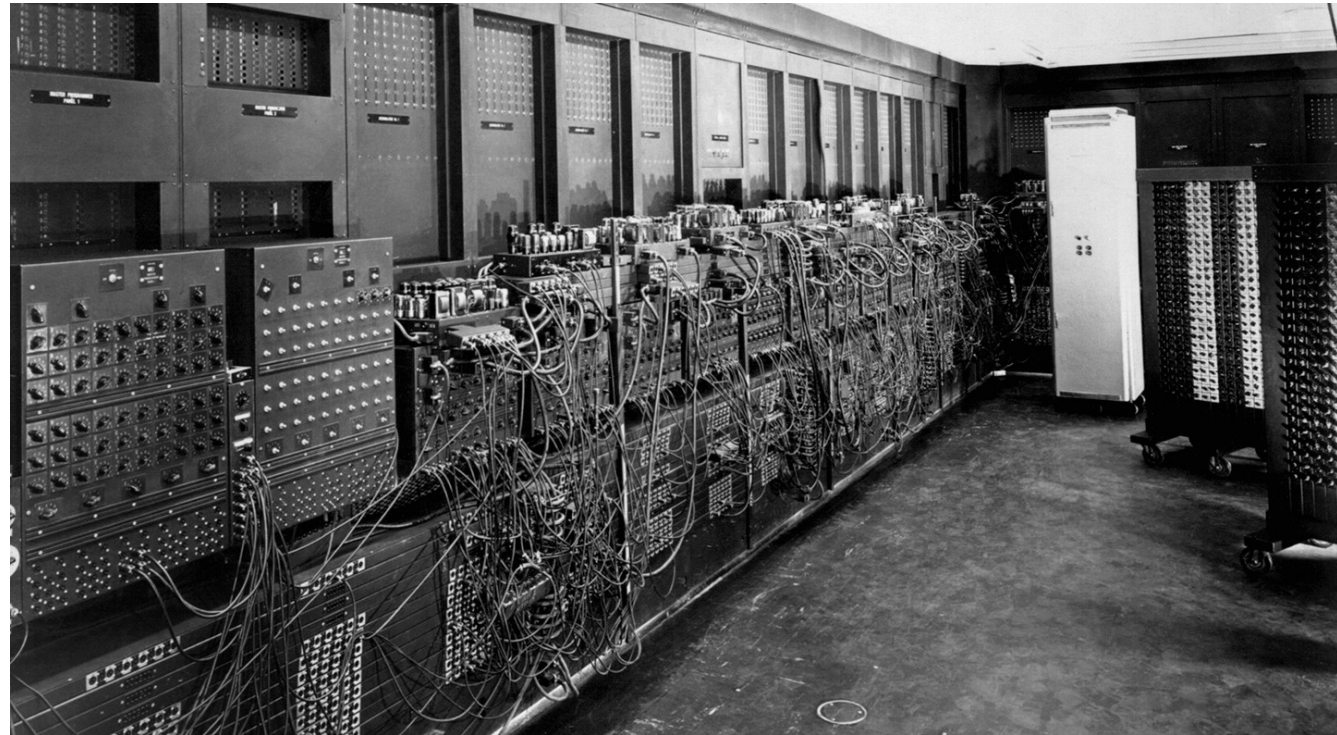
Post-Tech

➤ Digital Age (1950s–2000s):

Electricity, early computers, and personal computing transform industry and daily life.

➤ 21st Century & Post-AI Era:

AI, automation, and smart technologies accelerate innovation but increase energy use and e-waste.



ENIAC

Cost we pay



Do you know that while preparing this presentation, how much water and energy I wasted on my digital prompts and searches?

Responsible Future ??

➤ **Definition:**

Technology designed to meet present needs without compromising the future

➤ *Because every choice in technology today is a choice for the world we leave behind*





There is no planet 'B'...

~ Mike Berners



Why Sustainable Technology Matters now..

Exploding E-Waste Crisis

- ❖ In 2022, **62 million** tonnes of e-waste were generated globally, projected to rise to **82 million** tonnes by 2030.
- ❖ Only ~**22%** is formally recycled, creating supply chain vulnerabilities and toxic pollution risks.



From The Guardian

Carbon Footprint and Emissions

❖ Carbon Costs of Tech & AI:

The tech sector's emissions rival aviation, with AI systems alone projected to produce **~33–80 million tones of CO₂ by 2025**, often powered by fossil fuels.

❖ Professional Impact:

Rising emissions lead to **ESG regulations, investor scrutiny, and reputational risks** for companies and engineers.



oh mother earth...

Depleting Minerals for AI & Electronics

- ❖ AI systems and modern devices rely heavily on **lithium, cobalt, and rare earths**, with supply risks and projected shortages by 2030.

Professional Impact

- ❖ Engineers and companies face **higher costs, innovation delays, and pressure for ethical sourcing** in AI development.

Mother Earth by Diana Cheren Nygren



Moment to Reflect

Three Challenges, One Question

- ❖ E-waste, carbon emissions, and mineral depletion show that technology's **cost is of hidden, not absent**
- Can innovation still be called progress if it exhausts the planet that sustains it?





The Stone Age did not end because humans
ran out of stones. It ended because it was
time to rethink about how we live now.

~ William McDonough



Pathways to Sustainable Technology

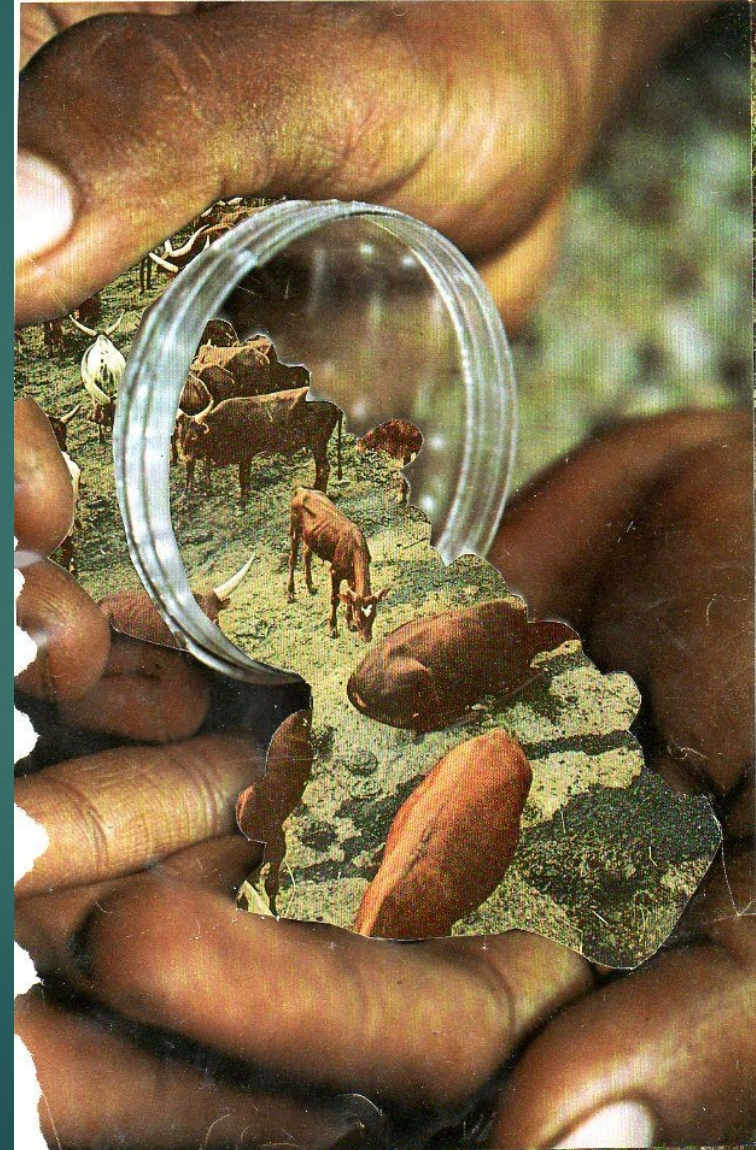
Reducing Digital Energy Waste

Green Coding: Reducing Digital Energy Waste:

- ▶ Optimized algorithms and carbon-aware computing can **cut compute energy use by 30–50%**

Professional Practice

- Developers **audit, refactor, and design software for efficiency**, following green software standards



Circular Economy & Modular Design

Extending Device Lifecycles

- ❖ Modular, repairable devices and recycled materials **reduce e-waste and resource extraction**

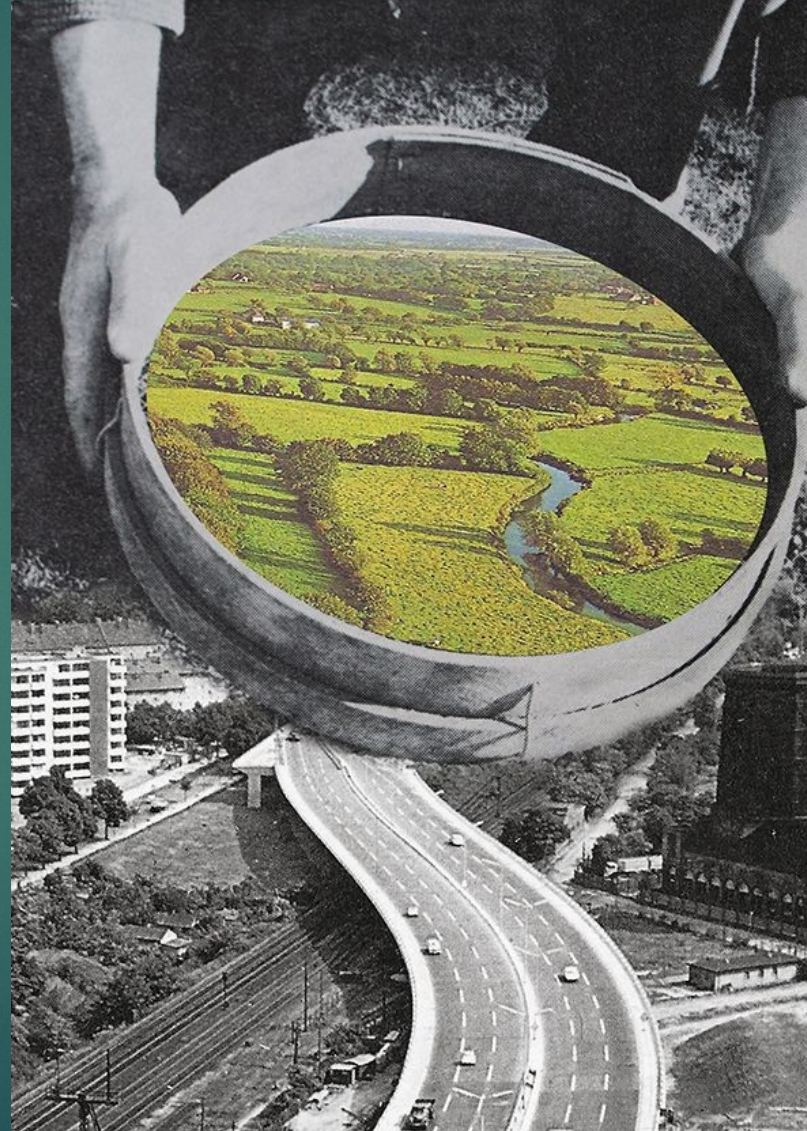
Professional Practice

- Engineers **design for longevity, repairability, and recyclability from day one**



Renewable Infrastructure & Responsible AI

- Renewable-powered data centers, low-power chips, and AI optimization **reduce emissions and energy waste**
- IT leaders **select sustainable providers and deploy efficient AI models**



Call to Action

Our Role as Technology Professionals

- ❖ Every design choice, line of code, and system decision **shapes environmental impact**

Way Forward

- ❖ Embed sustainability from the start — not as an afterthought...





“Let’s build technology that meets
today’s needs — and preserves
tomorrow’s world.”



Thank You